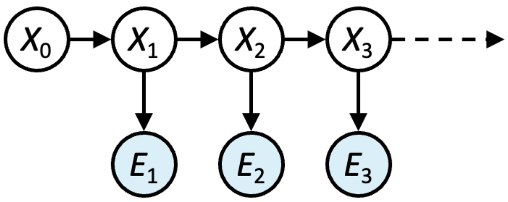
Hidden Markov Model

Practice Problems

QUESTION 1

Consider the following HMM, with hidden states X ∈ {a,b,c} and observations E ∈ {0,1}. Assume,

$$P(X_0) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



X_{t-1}	$P(X_t \mid X_{t-1})$		
	X_t		
	a	b	c
a	0.5	0.2	0.3
b	0.2	0.8	0
c	0.2	0.1	0.7

X_t	$P(E_t \mid X_t)$	
	E_t	
	0	1
a	0	1
b	1	0
c	0	1

You observe the sequence $\boldsymbol{E_1 = 0, E_2 = 1, E_3 = 1}$. Find the filtering probability at t=3, i.e. find $\boldsymbol{P(X_3 \mid E_1 = 0, E_2 = 1, E_3 = 1)}$.

Remember the filtering probability is given by the following equation.

$$\underbrace{P(X_t \mid e_{1:t})}_{\text{new estimate}} = \alpha \underbrace{P(e_t \mid X_t)}_{\text{sensor model}} \underbrace{\sum_{X_{t-1}} \underbrace{P(X_t \mid X_{t-1})}_{\text{transition model}} \underbrace{P(X_{t-1} \mid e_{1:t-1})}_{\text{previous estimate}}}_{\text{Prediction}}$$

Solution

Day 0 $\quad P(X_0) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Day 1

$$P(X_1) = \begin{bmatrix} 0.5 \\ 0.2 \\ 0.3 \end{bmatrix}(0) + \begin{bmatrix} 0.2 \\ 0.8 \\ 0 \end{bmatrix}(0) + \begin{bmatrix} 0.2 \\ 0.1 \\ 0.7 \end{bmatrix}(1) = \begin{bmatrix} 0.2 \\ 0.1 \\ 0.7 \end{bmatrix}$$

$$P(X_1|0) = \alpha \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0.2 \\ 0.1 \\ 0.7 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Day 2

$$P(X_2|0) = \begin{bmatrix} 0.2 \\ 0.8 \\ 0 \end{bmatrix}$$

$$P(X_2|0,1) = \alpha \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0.2 \\ 0.8 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

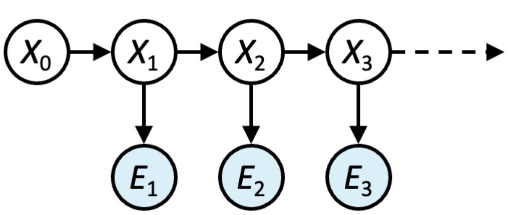
Day 3

$$P(X_3|0,1) = \begin{bmatrix} 0.5 \\ 0.2 \\ 0.3 \end{bmatrix}$$

$$P(X_3|0,1,1) = \alpha \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.2 \\ 0.3 \end{bmatrix} = \alpha \begin{bmatrix} 0.5 \\ 0 \\ 0.3 \end{bmatrix} = \begin{bmatrix} 0.65 \\ 0 \\ 0.38 \end{bmatrix}$$

Question 2

1. You sometimes get colds, which make you sneeze. You also get allergies, which make you sneeze. Sometimes you are well, which doesn't make you sneeze. You decide to model the process using the following HMM, with hidden states X ∈ {well, allergy, cold} and observations E ∈ {sneeze, quiet}. Assume, on day 0, you start well.



X_{t-1}	$P(X_t \mid X_{t-1})$
	X_t

	Well	Allergy	Cold
Well	0.7	0.2	0.1
Allergy	0.6	0.3	0.1
Cold	0.2	0.2	0.6

X_{t-1}	$P(E_t\|X_t)$	
	E_t	
	Quiet	Sneeze
Well	1	0
Allergy	0	1
Cold	0	1

You observe the sequence $\boldsymbol{E_1 = quiet, E_2 = sneeze, E_3 = sneeze}$. Find the filtering probability on day 3, i.e. find $\boldsymbol{P(X_3|E_1 = quiet, E_2 = sneeze, E_3 = sneeze)}$.

Remember the filtering probability is given by the following equation.

$$\underbrace{P(X_t|e_{1:t})}_{\text{new estimate}} = \alpha \underbrace{P(e_t|X_t)}_{\text{sensor model}} \underbrace{\sum_{X_{t-1}} \underbrace{P(X_t|X_{t-1})}_{\text{transition model}} \underbrace{P(X_{t-1}|e_{1:t-1})}_{\text{previous estimate}}}_{\text{Prediction}}$$

Your Solution: (you would not need to do any heavy calculation)

Day 0: You start well on day 0

$$P(X_o) = \begin{bmatrix} P(well) \\ P(allergy) \\ P(cold) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Day 1:

$$P(X_1) = \begin{bmatrix}0.7\\0.2\\0.1\end{bmatrix}(1) + \begin{bmatrix}0.6\\0.3\\0.1\end{bmatrix}(0) + \begin{bmatrix}0.2\\0.2\\0.6\end{bmatrix}(0)$$

$$= \begin{bmatrix}0.7\\0.2\\0.1\end{bmatrix}$$

$$P(X_1 \mid Quiet) = \alpha \begin{bmatrix}1\\0\\0\end{bmatrix}\begin{bmatrix}0.7\\0.2\\0.1\end{bmatrix} = \begin{bmatrix}1\\0\\0\end{bmatrix}$$

Day 2:

$$P(X_2 \mid Quiet) = \begin{bmatrix}0.7\\0.2\\0.1\end{bmatrix}$$

$$P(X_2 \mid Quiet, sneeze) = \alpha \begin{bmatrix}0\\1\\1\end{bmatrix}\begin{bmatrix}0.7\\0.2\\0.1\end{bmatrix} = \alpha \begin{bmatrix}0\\.2\\.1\end{bmatrix}$$

$$= \begin{bmatrix}0\\2/3\\1/3\end{bmatrix}, \quad \alpha = \frac{10}{3}$$

Day 3:

$$P(X_3 \mid Quiet, sneeze) = \begin{bmatrix}0.7\\0.2\\0.1\end{bmatrix}(0) + \begin{bmatrix}0.6\\0.3\\0.1\end{bmatrix}\left(\tfrac{2}{3}\right) + \begin{bmatrix}0.2\\0.2\\0.6\end{bmatrix}\left(\tfrac{1}{3}\right) = \begin{bmatrix}7/15\\4/15\\4/15\end{bmatrix}$$

$$P(X_3 \mid Quiet, sneeze, sneeze) = \alpha \begin{bmatrix}0\\1\\1\end{bmatrix}\begin{bmatrix}7/15\\4/15\\4/15\end{bmatrix}$$

$$= \begin{bmatrix}0\\1/5\\1/5\end{bmatrix}$$